

Looqbox Technical Challenge

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1. SQL Questions

a. What are the 10 most expensive products in the company?

SQL Question 1

```
query = """
SELECT
    PRODUCT_COD,
    PRODUCT_NAME,
    PRODUCT_VAL
FROM data_product
ORDER BY PRODUCT_VAL DESC
LIMIT 10;
"""

top10_produtos = pd.read_sql(query, conn)

top10_produtos
```

	PRODUCT_COD	PRODUCT_NAME	PRODUCT_VAL
0	301409	Whisky Escoces THE MACALLAN Ruby Garrafa 700mL	741.99
1	176185	Whisky Escoces JOHNNIE WALKER Blue Label Garra...	735.90
2	315481	Cafeteira Espresso 3 CORACOES Tres Modo Vermelho	499.00
3	100280	Vinho Portugues Tinto Vintage QUINTA DO CRASTO...	445.90
4	320046	Escova Dental Eletica ORAL B D34 Professional...	399.90
5	190817	Champagne Rose VEUVE CLICQUOT PONECARDIM Garraf...	366.90
6	153795	Champagne Frances Brut Imperial MOET Rose Gar...	359.90
7	311397	Conjunto de Panelas Alegria em Inox TRAMONTINA...	359.00
8	147706	Whisky Escoces CHIVAS REGAL 18 Anos Garrafa 750ml	329.90
9	154431	Champagne Frances Brut Imperial MOET & CHANDON...	315.90

b. What sections do the 'BEBIDAS' and 'PADARIA' departments have?

SQL Question 2

```
query = """
SELECT DISTINCT
    DEP_NAME,
    SECTION_NAME
FROM data_product
WHERE DEP_NAME IN ('BEBIDAS', 'PADARIA')
ORDER BY DEP_NAME, SECTION_NAME;
"""

resultado = pd.read_sql(query, conn)

print(resultado)
```

	DEP_NAME	SECTION_NAME
0	BEBIDAS	BEBIDAS
1	BEBIDAS	CERVEJAS
2	BEBIDAS	REFRESCOS
3	BEBIDAS	VINHOS
4	PADARIA	DOCES-E-SOBRREMESAS
5	PADARIA	GESTATE
6	PADARIA	PADARIA
7	PADARIA	QUEIJSOS-E-FRIOS

c. What was the total sale of products (in \$) of each Business Area in the first quarter of 2019?

```
SQL Question 3

Query = """
SELECT
    sc.BUSINESS_NAME,
    SUM(ps.SALES_VALUE) AS TOTAL_SALES
FROM data_product_sales ps
INNER JOIN data_store_cad sc
    ON ps.STORE_CODE = sc.STORE_CODE
WHERE ps.DATE BETWEEN '2019-01-01' AND '2019-03-31'
GROUP BY sc.BUSINESS_NAME
ORDER BY TOTAL_SALES DESC;
"""

q3 = pd.read_sql(query, conn)

q3["TOTAL_SALES"] = q3["TOTAL_SALES"].map(lambda x: f"R$ {x:,.2f}")

print(q3)

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C:\Users\Caroline Nicolau\AppData\Local\Temp\ipykernel_12788\4886343085.py:13: UserWarning: pandas only supports SQLAlchemy connectable (engine/connection) or database string URI or sqlite3 DBAPI2 connection. Other DBAPI2 objects are not tested. Please consider using SQLAlchemy.
q3 = pd.read_sql(query, conn)

BUSINESS_NAME    TOTAL_SALES
0      Farma      R$ 81,776,691.73
1      Varelo      R$ 81,487,347.45
2      Atacado      R$ 80,384,884.68
3      Proximidade  R$ 80,171,122.88
4      Posto       R$ 32,072,326.48
```

2. Cases

a. Answer Case 1

```
Case 1

from Functions import retrieve_data

dados = retrieve_data(
    product_code=18,
    store_code=1,
    date=["2019-01-01", "2019-01-31"]
)

dados["SALES_VALUE"] = dados["SALES_VALUE"].map(lambda x: f"R$ {x:,.2f}")

dados.head()

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C:\Users\Caroline Nicolau\Documents\Longbox_Challenge\Functions.py:42: UserWarning: pandas only supports SQLAlchemy connectable (engine/connection) or database string URI or sqlite3 DBAPI2 connection. Other DBAPI2 objects are not tested. Please consider using SQLAlchemy.
df = pd.read_sql(query, conn)

STORE_CODE  PRODUCT_CODE  DATE  SALES_VALUE  SALES_QTY
0           1           18  2019-01-01      R$ 708.50        65.0
1           1           18  2019-01-02      R$ 1,227.10       119.0
2           1           18  2019-01-03      R$ 1,644.50       105.0
3           1           18  2019-01-04      R$ 1,000.00       100.0
4           1           18  2019-01-05      R$ 893.80         82.0
```

b. Answer Case 2

```
Case 2

conn = pymysql.connect(
    host="19.188.110.114",
    user="longbox-challenge",
    password="long-challenge",
    database="longbox-challenge",
    port=3306
)

query1 = """
SELECT
    STORE_CODE,
    STORE_NAME,
    START_DATE,
    END_DATE,
    BUSINESS_NAME,
    BUSINESS_CODE
FROM data_store_cad
"""

query2 = """
SELECT
    STORE_CODE,
    DATE,
    SALES_VALUE,
    SALES_QTY
FROM data_store_sales
WHERE DATE BETWEEN '2019-01-01' AND '2019-12-31'
"""

df_store = pd.read_sql(query1, conn)
df_sales = pd.read_sql(query2, conn)

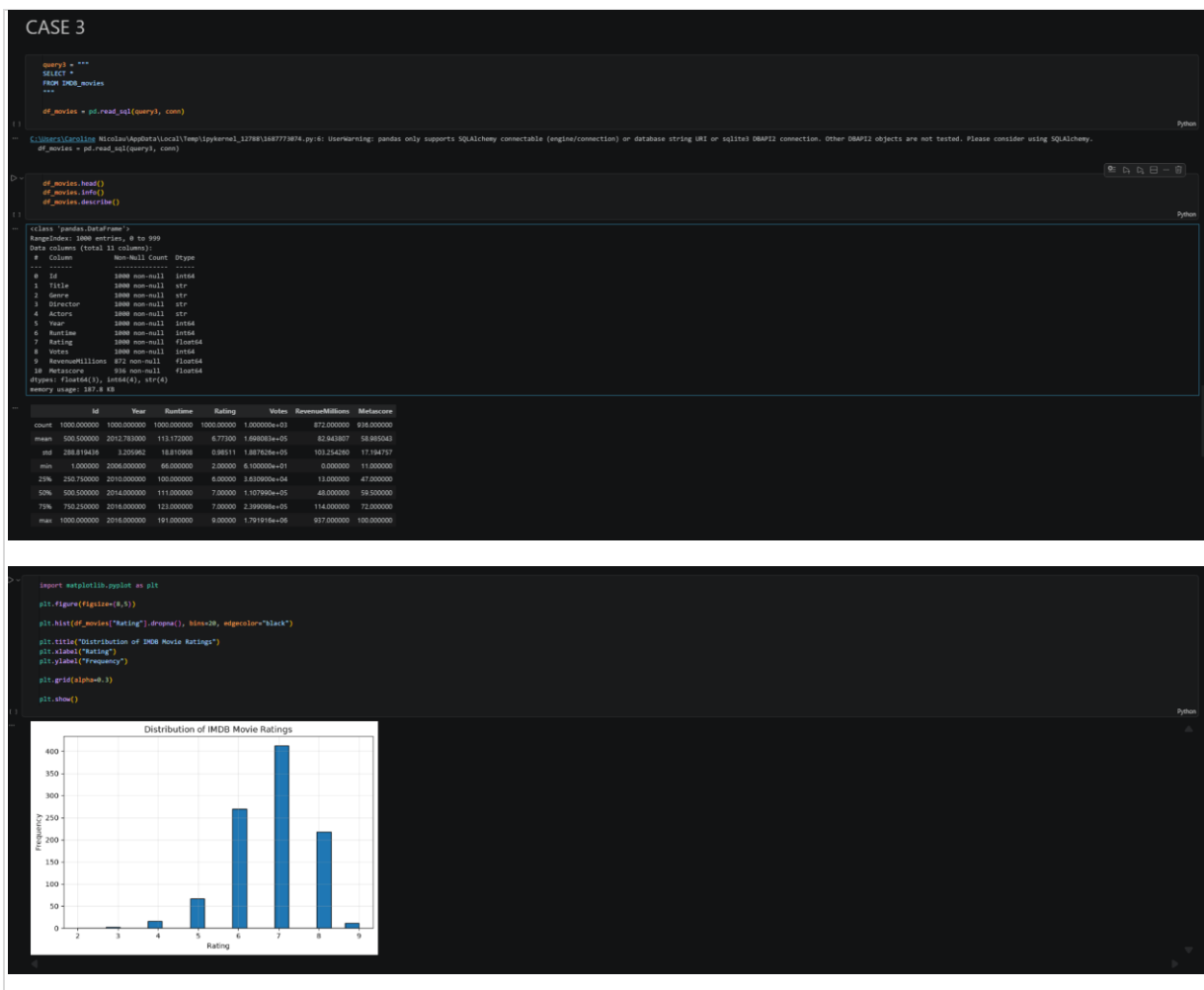
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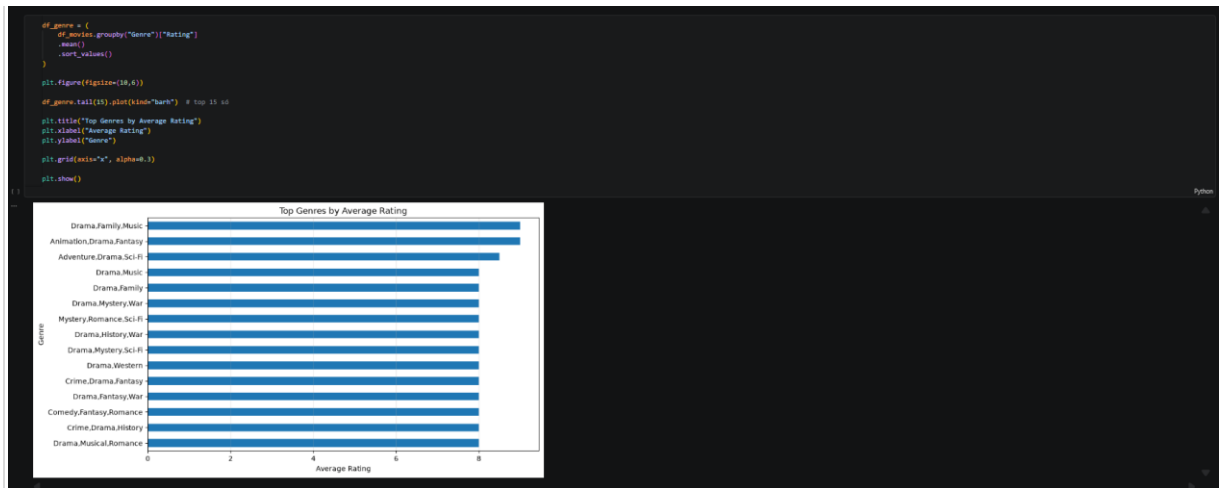
C:\Users\Caroline Nicolau\AppData\Local\Temp\ipykernel_4248\3883824289.py:38: UserWarning: pandas only supports SQLAlchemy connectable (engine/connection) or database string URI or sqlite3 DBAPI2 connection. Other DBAPI2 objects are not tested. Please consider using SQLAlchemy.
df_store = pd.read_sql(query1, conn)

C:\Users\Caroline Nicolau\AppData\Local\Temp\ipykernel_4248\3883824289.py:31: UserWarning: pandas only supports SQLAlchemy connectable (engine/connection) or database string URI or sqlite3 DBAPI2 connection. Other DBAPI2 objects are not tested. Please consider using SQLAlchemy.
df_sales = pd.read_sql(query2, conn)
```



c. Answer Case 3





Bônus — Streamlit Application (app.py)

Escolha uma análise

Explorar schema

Deploy

Looqbox Data Explorer

Tabelas disponíveis no banco

Tables_in_looqbox-challenge	
0	IMDB_movies
1	data_product
2	data_product_sales
3	data_store_cad
4	data_store_sales

Tela inicial — Explorar schema

Escolha uma análise

Produtos mais caros

Deploy

Looqbox Data Explorer

Top 10 produtos mais caros

	PRODUCT_COD	PRODUCT_NAME	PRODUCT_VAL	DEP_NAME	DEP_COD	SECTION_NAME
0	301409	Whisky Escoces THE MACALLAN Ruby Garrafa 700ml com Caixa	741.99	BEBIDAS	2	BEBIDAS
1	176185	Whisky Escoces JOHNNIE WALKER Blue Label Garrafa 750ml	735.9	BEBIDAS	2	BEBIDAS
2	315481	Cafeteira Expresso 3 CORACOES Tres Modo Vermelho	499	BEBIDAS	2	BEBIDAS
3	100280	Vinho Portugues Tinto Vintage QUINTA DO CRASTO Garrafa 750ml	445.9	BEBIDAS	2	VINHOS
4	320046	Escova Dental Eletrica ORAL B D34 Professional Care 5000 110v	399.9	PERFUMARIA	8	HIGIENE BUCAL
5	190817	Champagne Rose VELVE CLICQUOT PONSARDIM Garrafa 750ml	366.9	MERCEARIA	6	ARTIGOS-PARA-O-LAR
6	153795	Champagne Frances Brut Imperial MOET Rose Garrafa 750ml	359.9	MERCEARIA	6	ARTIGOS-PARA-O-LAR
7	311397	Conjunto de Panelas Alegria em Inox TRAMONTINA 5 Pecas Gratis Utensilios 5 Pecas	359	MERCEARIA	6	ARTIGOS-PARA-O-LAR
8	147706	Whisky Escoces CHIVAS REGAL 18 Anos Garrafa 750ml	329.9	BEBIDAS	2	BEBIDAS
9	154431	Champagne Frances Brut Imperial MOET & CHANDON Garrafa 750ml	315.9	MERCEARIA	6	ARTIGOS-PARA-O-LAR

<<

Escolha uma análise

Seções por departamento

Deploy

Looqbox Data Explorer

Seções de BEBIDAS e PADARIA

	DEP_NAME	SECTION_NAME
0	BEBIDAS	BEBIDAS
1	BEBIDAS	VINHOS
2	PADARIA	DOCES-E-SOBREMESAS
3	PADARIA	QUEIJOS-E-FRIOS
4	BEBIDAS	CERVEJAS
5	PADARIA	PADARIA
6	BEBIDAS	REFRESCOS
7	PADARIA	GESTANTE

<<

Escolha uma análise

Vendas Q1 2019

Deploy

Looqbox Data Explorer

Total de vendas por Business Area (Q1 2019)

	BUSINESS_NAME	TOTAL_SALES
0	Farma	81776691.73
1	Varejo	81032347.65
2	Atacado	80384884.6
3	Proximidade	80171122.8
4	Posto	32072326.4

